

FURTHER EVIDENCE FOR A FRÖHLICH INTERACTION OF ERYTHROCYTES

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The aggregation of red blood cells, of a single species and of mixtures of species, deviates from brownian aggregation theory. The deviations are compatible with Fröhlich's theory of a quantum mechanical interaction between living cell membranes.

In a previous paper [1] we showed that a newly discovered attraction between human erythrocytes was compatible with a theory of Fröhlich [2,3] of long-range coherent interactions in living cells. As necessary in the theory the attraction disappeared when the cellular membrane was disorganised (with glutaraldehyde) and disappeared reversibly when the transmembrane potential difference was abolished or when the cells were depleted of their supply of metabolic energy. The interaction is associated with subsequent aggregation or rouleau formation [4–7] and does not occur unless there are rouleau-inducing macromolecules in the suspending medium. In place of the simple geometrical analysis previously used we now employ an adaptation of the Smoluchowski theory of colloid aggregation by brownian movement [8]. This method gives an estimate of the range of the interaction and provides evidence that the interaction is specific, from experiments on the aggregation of mixtures of erythrocytes from different mammalian species.

Chandrasekhar [9] reviewed Smoluchowski's theory and gives an expression for the total number of particles (aggregates and singles) N_t , in a coagulating colloid suspension as a function of time t [9, eq. (461)]

$$N_t^{-1} - N_0^{-1} = (4kTZ/3\eta r) t, \quad (1)$$

where η is the viscosity of the suspending medium, r the average radius of the particles and Z the radius of the hypothetical zone of attraction around each particle. In our experiments the brownian motion of erythro-

cytes in their own plasma, resting on the floor of a sealed siliconized haemocytometer chamber, is recorded by time-lapse cinémicroscopy [1]. When cells touch they quickly slide together [4,6,7] face to face. So linear aggregates, called rouleaux, form in time in the chamber (fig. 1). The process is similar to the coagulation of a colloid though the "particles" are much larger and the aggregates more regular. Eq. (1) would apply if the cells had the same density as the suspending fluid and so were distributed randomly in depth as well as in the horizontal extent of the chamber. In fact the cells have a slightly higher density and quickly fall to the floor. The three-dimensional process described by eq. (1) is thus compressed into two dimensions. The number of collisions in brownian motion and so the rate constant are accordingly increased by a factor of $3/2$ giving

$$N_t^{-1} - N_0^{-1} = (2kTZ/\eta r) t. \quad (2)$$

Einstein's theory of brownian motion [10] gives the mean square deviation in two dimensions as a function of time,

$$\langle U^2 \rangle = 4kTt\mu^{-1}, \quad (3)$$

where μ is Stokes' drag coefficient for spheres,

$$\mu = 6\pi\eta r. \quad (4)$$

To check that cells beyond the range of an interaction were executing brownian motion, U was measured for each of 30 cells which remained further than $6 \mu\text{m}$

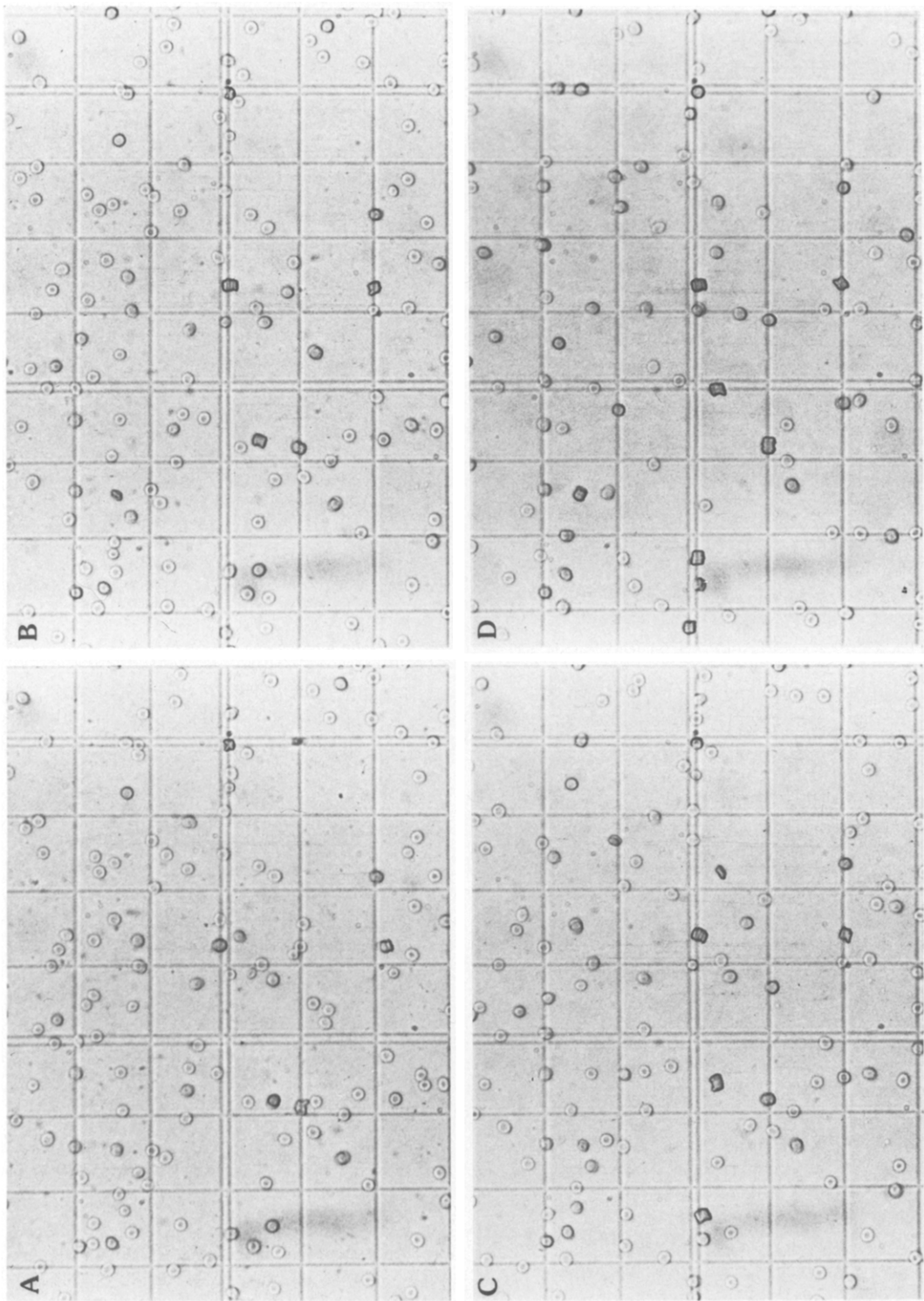


Fig. 1. Normal cells in plasma. Four frames from a ciné film. (A) the cells have just settled and some rouleaux have formed; (B) after 6 min; (C) after 22 min; (D) after 114 min.

Table 1
Drag coefficients.

	$\mu \times 10^7 \text{ (N s m}^{-1}\text{)}$ experimental value	experim. value/ Stokes' value, oblate spheroids
normal cells	$2.80 \pm 0.66^a)$	$1.82 \pm 0.43^a)$
cells fixed in glutaraldehyde	2.96 ± 0.68	1.92 ± 0.44
cells at $pH = 6.3$	2.78 ± 0.76	1.80 ± 0.50
cells restored to $pH \approx 7.6$	2.57 ± 0.83	1.67 ± 0.54
ionophore-treated cells	3.39 ± 0.14	2.21 ± 0.09
metabolically depleted cells	2.18 ± 0.34	1.42 ± 0.23
cells repleted with adenosine	2.68 ± 0.74	1.74 ± 0.48
mean value for cells	$2.75 \pm 0.10^b)$	$1.78 \pm 0.06^b)$
microspheres $5.8 \mu\text{m}$	$1.06^c)$	1.8
microspheres $8.0 \mu\text{m}$	$1.67^c)$	2.1

a) Standard deviation. b) Standard error of the mean.

c) Microspheres in plasma stick to the floor of the chamber. These experiments were done in soapy water ($\eta = 1.06 \times 10^{-3} \text{ Pa s}$).

from other cells during the first 15 min of each experiment. Beyond 15 min fewer than 30 cells remained unassociated. For an average plasma viscosity ($1.7 \times 10^{-3} \text{ Pa s}$) and incorporating Perrin's correction to eq. (4) for oblate spheroids [11] the calculated value for $\mu = 1.54 \times 10^{-7} \text{ N s m}^{-1}$. The experimental values for cells (table 1) are nearly twice as large and a similar discrepancy occurs with microspheres (table 1). We infer that the hydrodynamic drag is substantially increased when cells or microspheres are resting on the floor of the haemocytometer. In deriving eqs. (1) and (2) it is assumed [9] that the diffusion coefficient of the individual particles is given by the Stokes-Einstein relation

$$D = kT/6\pi\eta r, \quad (5)$$

where the Stokes' drag coefficient appears as the denominator. The rate constant in eq. (2) must therefore be reduced by the ratio of the experimentally determined drag to the Stokes' drag, and this ratio is determined for each experiment. In seven experiments conducted on the same blood the standard error of the mean value of $\langle U^2 \rangle$, and so of μ was 19%. This sets a lower limit for the accuracy of the experiments as presently performed.

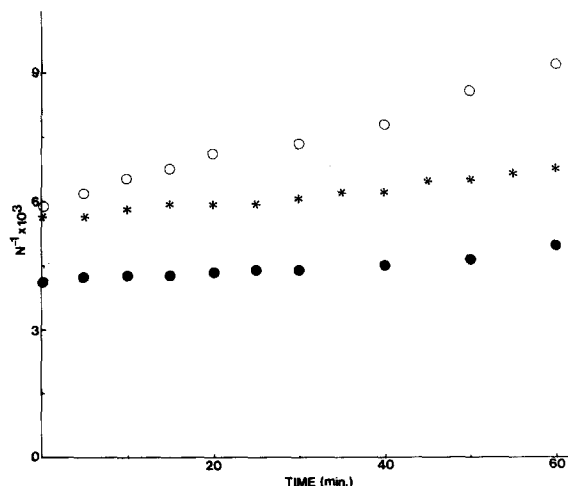


Fig. 2. Plots of the reciprocal of the total number of particles in an area of $10^5 \mu\text{m}^2$ against time. $\circ$ normal cells, $\Xi = 3.37$; $\bullet$ glutaraldehyde-fixed cells, $\Xi = 0.83$; $*$ microspheres, $\Xi = 1.15$.

For a typical experiment and if there is no interaction before cell contact ($Z = r$) the calculated corrected slope of the graph of N_t^{-1} against t will be $2.6 \times 10^{-18} \text{ m}^3 \text{ s}^{-1}$, a value which varies slightly from experiment to experiment. Experimental results are displayed in figs. 2-4 and in column 3 of table 2 as multiples of this base value (interaction coefficient $\Xi = Z/r$). In this table results obtained by the previous method [1] are displayed

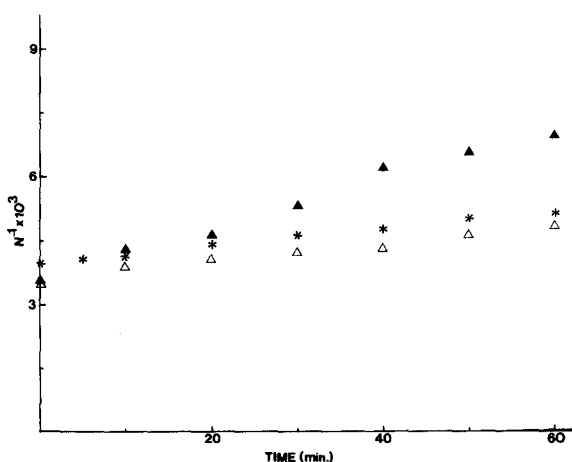


Fig. 3. As in fig. 2. Δ membrane potential reduced to 1.5 mV by changing pH of plasma to 6.3, $\Xi = 1.28$; $\blacktriangle$ membrane potential restored by returning pH to 7.6, $\Xi = 3.51$; $*$ membrane potential reduced by treatment of cells with ionophore A23187, $\Xi = 1.66$.

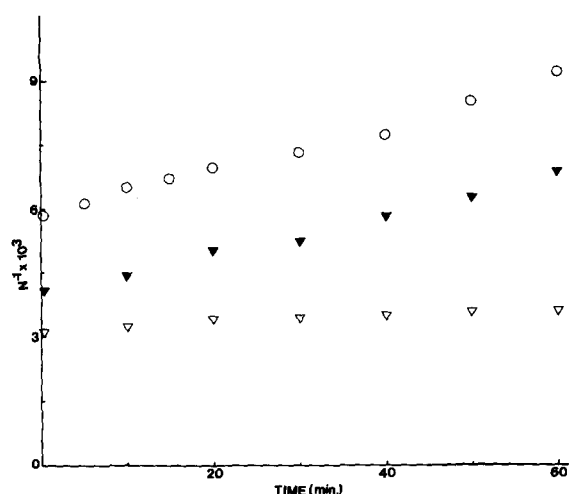


Fig. 4. As in figs. 2 and 3. ∇ cells depleted of metabolic energy, $\Xi = 0.40$; $\blacktriangledown$ cells repleted with adenosine, $\Xi = 3.65$; $\circ$ normal cells, $\Xi = 3.37$.

alongside (column 2). By having the same trend the results confirm each other though the relationship between the two sets of figures is non-linear as may be expected.

Fixed cells and microspheres have interaction coefficients close to unity and so, taking account of the added drag, show no significant deviation from brownian move-

ment. For normal cells the interaction coefficient is three times the base value. Abolishing the membrane potential by reducing pH to 6.3 abolishes the interaction. Abolishing the membrane potential by the ionophore causes a significant decrease in Ξ but the two experiments suggest a remnant of interaction. Metabolic depletion causes a great fall in Ξ to a value which suggests a repulsion but both of these two last findings need further study for confirmation. Repleting energy-depleted cells and partially restoring the membrane potential after 3 h [1] brings the interaction back to the range of normal cells. Since the interaction coefficient $\Xi = Z/r$ we conclude that the interaction of normal cells extends on the average to a distance of three cell radii.

An assumption of the Smoluchowski theory of colloid aggregation is that all the aggregating particles are of the same size which is a reasonable assumption for the erythrocytes from a single individual. Mixtures of erythrocytes from different species of mammals will also aggregate into rouleaux when suspended in isotonic solutions of certain macromolecules [12]. Plasma cannot be used because of immunological reactions. In such interspecies rouleaux, cell volumes may differ by as much as a factor of two. Overbeek [13], in discussing the aggregation of polydispersed colloids, shows that

Table 2
Interaction coefficients Ξ .

	Probability a)	Ξ b)	No. of exp.	No. of subjects
normal cells	0.85	2.92 ± 1.12 c)	15	9
cells fixed in glutaraldehyde	0.46	0.92 ± 0.46	9	5
cells at $pH = 6.3$	0.40	1.07 ± 0.41	10	5
cells restored to $pH \approx 7.6$	0.71	2.57 ± 0.99	9	3
ionophore-treated cells	0.48	1.69 ± 0.04	2	2
metabolically depleted cells	0.44	0.43 ± 0.11	4	2
cells repleted with adenosine	0.87	3.26 ± 0.68	3	2
microspheres $5.9 \mu m$		0.93	1	
microspheres $8.0 \mu m$		1.15	1	

a) Probability of pairs of cells colliding in 3 h [1]. Value for no interaction ≈ 0.3 .

b) Interaction coefficient as defined in text. $\Xi > 1$ attraction.

c) Standard deviation: $p < 0.001$ normal versus glutaraldehyde cells; normal versus $pH = 6.3$ cells; $pH = 6.3$ versus $pH = 7.6$ cells; normal versus metabolically depleted cells; $p < 0.003$ normal versus ionophore-treated cells.

the *least* likely collision is between particles of the same size. Yet in the interspecies experiments [12] cells of the *same* size and so of the same species show a preference to adhere to each other. In ten combinations of different species the only exception was a rat-mouse mixture though there is a significant difference in cell size, larger than in some pairs which did show a species preference. Perhaps rats and mice, being of the same biological family, have red cells indistinguishable in respect to a Fröhlich interaction.

The interspecies interaction is contrary to that expected of a passive phenomenon [13]. It gives a different type of evidence from that adduced above for a specific interaction at a distance between mammalian erythrocytes. Further work will reveal whether the interspecies interaction is a repulsion, a null interaction, or an attraction smaller than that between cells of the same species. Brownian motion could overcome a small repulsion allowing rouleaux to form at a slow rate.

During growth, particularly in the embryo, cells appear to have the ability to recognise and to adhere to like cells. Experimentally a random mixture of dispersed cells of embryonic liver and heart will sort itself out. The hepatocytes form a ring around, and an island within, the cardiac myocytes [14]. In synchronized Chinese Hamster Ovary-KI cells, GI cells adhere to each other at twice the rate S cells adhere, while GI cells adhere to S cells at an intermediate rate [15]. In explanation, cell motility and chemical and physicochemical mechanisms are invoked [16]. Mammalian erythrocytes are not motile so their mutual attraction is not chemotactic. The first time they touch they adhere and then quickly slide into the rouleau array [4]. So the process is not one of tentative random collisions followed by immobilisation as chemical binding sites interlock. Yet their interactions are not dissimilar from those reported for other groups

of cells [14-16]. This suggests that a Fröhlich effect is playing at least some part in the interaction of a wide range of mammalian cells.

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